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# Control of energy homeostasis by insulin and leptin: Targeting the arcuate nucleus and beyond

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ABSTRACT

As the obesity epidemic, diabetes mellitus type 2, and associated comorbidities show no signs of abating, large efforts have been put into a better understanding of the homeostatic control mechanisms involved in regulation of body weight and energy homeostasis. For decades, the hypothalamic arcuate nucleus (ARC), which integrates peripheral signals and modulates appetite and metabolism, has been the focus of investigation. Besides these basic homeostatic circuits, food palatability and reward are thought to be major factors involved in the regulation of food intake. Highly palatable food is easily available, and is ingested even when there is no metabolic need for it. Thus, overriding of the homeostatic control systems by the cognitive, rewarding, social, and emotional aspects of palatable food may contribute to the obesity epidemic. This review aims to provide an updated view, how insulin and leptin as signals originating from the periphery of the body and communicating energy availability to the CNS act not only on ARC neurons, but also directly control the activity of neuronal circuits in control of food-associated reward mechanisms.

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Already in the 1950s, Kennedy put forth a homeostatic model for the control of energy homeostasis and body weight [1]. According to this model the release of signaling mediators from peripheral tissues would communicate to the central nervous system (CNS) the availability of fuel sources, thus allowing for the appropriate adaptation of responses such as food intake and energy expenditure. While for the longest time the exact nature of these mediators remained elusive, the recent identification that peripheral hormones such as leptin, insulin, and numerous others, as well as nutrients directly can communicate to the hypothalamus to control energy homeostasis, has set the ground for detailed understanding of the neuronal circuitry underlying the control of body weight homeostasis. Nevertheless, a growing body of evidence indicates that this basic homeostatic type of regulation – mainly mediated through the hypothalamus – relies on a complex intra-hypothalamic neuronal circuitry but also can be overruled by higher brain functions influenced by the addictive value of palatable food as well as its visual and gustatory aspects. Thus, this review will focus on the integrative view of the hormonal control of not only hypothalamic neuronal circuits but also further reaching circuits to ultimately determine the complex behavior underlying energy homeostasis.

## 1. Insulin and leptin as adiposity signals

The finding that insulin, when infused into the brain, decreases food intake and body weight led to the proposal by Woods and Porte that insulin serves as an adiposity signal in the brain. Here, it provides a signal related to the amount of body fat and causes a long-term, net catabolic response, decreasing food intake and increasing energy expenditure [2–5]. Importantly, insulin reaches the brain not only under these experimental conditions, but has access to the brain via a saturable transporter across the blood brain barrier [6]. Insulin is secreted acutely in response to an increase in blood glucose, e.g. after consumption of a meal, and plasma insulin levels are also directly correlated with the degree of body adiposity [7–9]. Both, in its peripheral target tissues (mainly the liver, skeletal muscle and adipose tissue), as well as the CNS, insulin binds to its receptor, which belongs to the family of tyrosine kinase receptors, thereby initiating an intracellular signaling cascade [10].

Binding of insulin leads to rapid autophosphorylation of the insulin receptor, followed by tyrosine phosphorylation of insulin receptor substrate (IRS) proteins, which subsequently activates downstream pathways such as the phosphatidylinositol 3 kinase (PI3 kinase) and the mitogen-activated protein kinase (MAPK) cascades [11] (Fig. 1).

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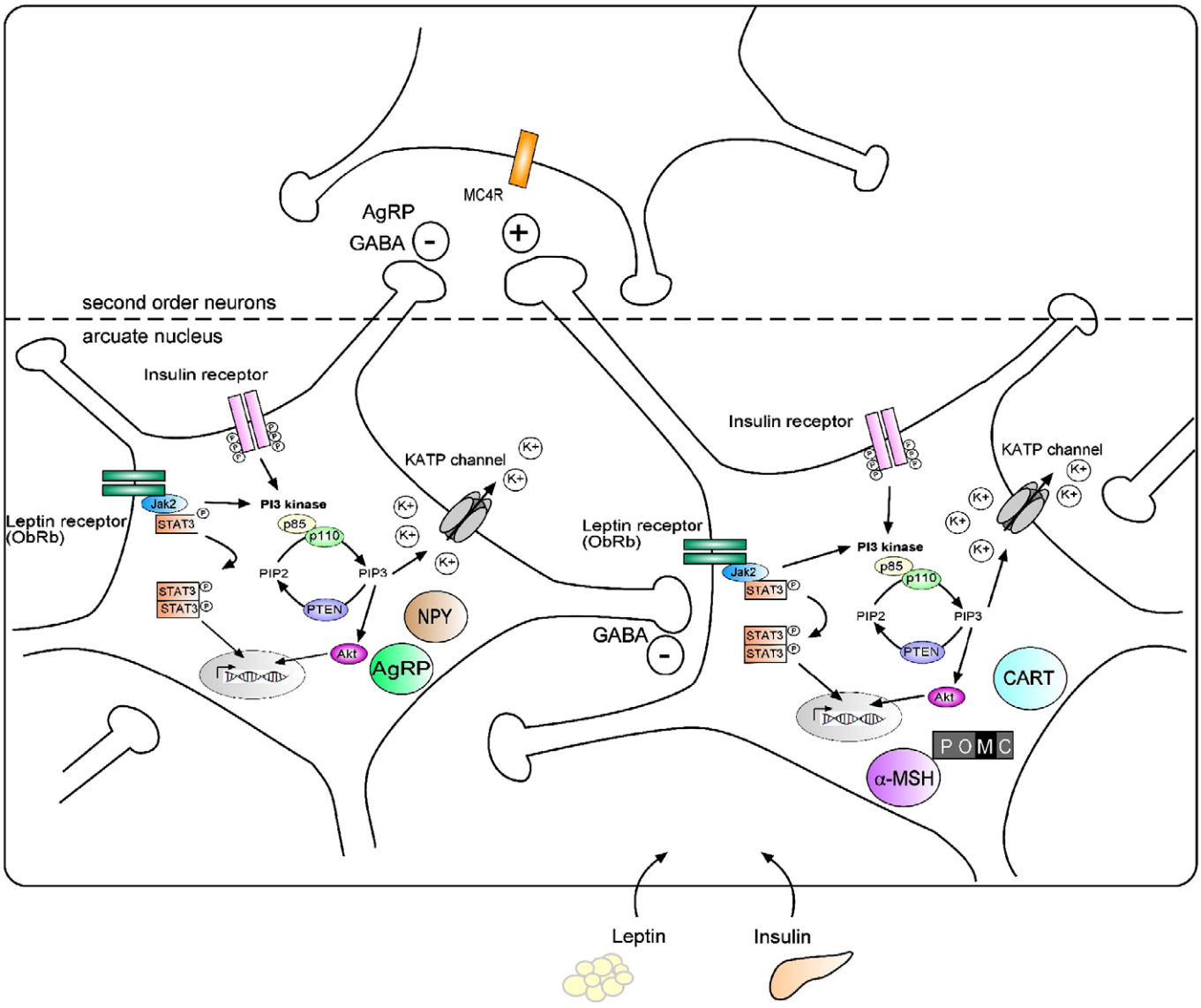



**Fig. 1.** Mechanisms of insulin and leptin action in the hypothalamus. Binding of insulin to its receptor initiates a signal transduction cascade resulting in PI3 kinase activation (p85: regulatory subunit; p110: catalytic subunit). PI3 kinase phosphorylates phosphatidylinositol 3,4-diphosphate (PIP2) to phosphatidylinositol 3,4,5-triphosphate (PIP3), which opens $K_{ATP}$ channels, thereby producing an outward flow of $K^+$ ions. This leads to hyperpolarization and reduced firing rate of the neuron. Activation of the leptin receptor (ObRb) produces relatively weak stimulation of PI3 kinase via IRS proteins but strong Stat3 phosphorylation via activation of Jak2. Insulin and leptin activate the melanocortin system, which comprises POMC/CART-expressing and AgRP/NPY-expressing neurons of the arcuate nucleus. Activation of POMC neurons triggers release of the anorexigenic hormone α-MSH at target sites in second order neurons, where it binds to melanocortin 4 receptor (MC4R). By binding to MC4R, AgRP acts as an inverse agonist of the melanocortin system, thereby antagonizing the effects of α-MSH. AgRP neurons also counteract the actions of melanocortin signaling by directly inhibiting POMC neurons, as well as postsynaptic neurons via GABA release. Abbreviations: AgRP: agouti-related peptide; α-MSH: alpha melanocyte-stimulating hormone; CART: cocaine and amphetamine related transcript; GABA: gamma-aminobutyric acid; NPY: neuropeptide Y; POMC: proopiomelanocortin; PTEN: phosphatase and tensin homolog.

Activation of the PI3 kinase results in activation of protein kinase B/Akt and phosphorylation of FOXO, which is of critical importance for maintenance of energy homeostasis by the CNS [12,13].

Among many others, the best-characterized biological actions of insulin include the stimulation of glucose transport in skeletal muscle and adipose tissue, as well as the suppression of hepatic glucose production [10]. However, insulin action in the CNS in control of energy homeostasis and peripheral glucose metabolism clearly involves other mechanisms, such as regulation of neuropeptide transcription and release [13].

Mice with a brain-specific insulin receptor deficiency are obese and display impaired fertility, assigning insulin receptor signaling a role in regulation of fuel metabolism and reproduction [14]. Moreover, multiple studies have shown that insulin signaling in the brain not only controls body weight, but also regulates peripheral glucose metabolism [15–17].

In 1994, another, by now the proteotypic adiposity signal, was discovered as the *ob* (obese) gene product, which encodes the leptin protein [18]. Leptin is mainly produced by adipocytes in the white adipose tissue (WAT) and released into the circulation in proportion to the body fat content [19,20]. Mice carrying a mutation in the *ob* gene exhibit dramatic weight gain (*ob*/*ob*) [18], and a naturally occurring loss-of-function mutation of the leptin receptor in both CNS and periphery underlies the phenotype of obese and diabetic mice (*db*/*db*) [21]. In humans, rarely occurring loss-of-function mutations of the leptin gene result in morbid obesity, that can be rescued by leptin administration [22,23]. As shown by Cohen et al., the neuron specific deletion of the leptin receptor resembles the phenotype of the *db*/*db* mouse [24], indicating that indeed the major role of leptin action to control energy homeostasis is accounted for its signaling in the CNS. Along this line, administration of leptin in the CNS decreases food intake and body weight in various species, underlining the pivotal role

of leptin in the regulation of energy homeostasis in the CNS [25]. Similarly to insulin, leptin uptake by the brain depends on a saturable transport across the blood brain barrier [6].

The leptin receptor is a cytokine receptor of which different isoforms, arising from alternative splicing, are expressed in the CNS [26]. Among these different isoforms only the long isoform (ObRb) has been shown to activate intracellular signaling [27]. Hormone binding to the receptor induces activation of the JAK/STAT pathway, but also activation of PI3 kinase signaling (Fig. 1). It has been demonstrated that the coordinated activation of STAT3 and phosphorylation of FOXO is required for leptin-evoked POMC transcription [28,29]. On the other hand, leptin's ability to control POMC-cell firing is mediated via yet unidentified cation channels [30] and also depends on PI3 kinase signaling [31]. The latter pathway thus provides a potential point of convergence and synergism with insulin-stimulated signals [32].

Receptors for both insulin and leptin are expressed in various regions of the brain [33–36] and extensive research over the last 30 years has identified the mediobasal hypothalamus, a region mainly involved in the regulation of energy homeostasis, as a major target for insulin and leptin action [37].

## 2. The melanocortin system in control of energy homeostasis

In the arcuate nucleus (ARC) of the mediobasal hypothalamus, two functionally opposing neuronal populations have been studied in great detail: the proopiomelanocortin/cocaine and amphetamine related transcript (POMC/CART)-expressing and the agouti-related peptide/neuropeptide Y (AgRP/NPY)-expressing neurons (Fig. 1). The anorectic POMC/CART neurons express POMC as a precursor protein which, dependent on the cell-type specific expression pattern of prohormone convertases, is processed to different bioactive products [38]. Among these are the melanocyte-stimulating hormones (α-, β-, and γ-MSH). α- and β-MSH reduce food intake and increase energy expenditure both in animals and humans [39–41]. α- and β-MSH act on melanocortin receptor (MC-R) types 3 and 4, which are expressed in the ARC, paraventricular nucleus (PVN), lateral hypothalamus (LH), ventromedial hypothalamus (VMH), and dorsomedial hypothalamus (DMH) [42,43]. Recent elegant cell-type specific reconstitution experiments of MC4-R on a null background in mice have defined single-minded homolog 1 (SIM1)-positive PVN neurons as the critical mediators of α-MSH stimulated anorexia [44]. However, since energy expenditure remained unaltered in PVN/SIM1-reconstituted MC4-R null mice, the nature of the energy expenditure controlling α-MSH target neurons remains elusive [44].

The second principal neuronal populations in the ARC are the orexigenic AgRP/NPY neurons. NPY is a potent stimulator of food intake and it reduces energy expenditure [45,46]. AgRP acts as an inverse agonist of the MC3/4-R and prevents the anorectic effect of α-MSH (Fig. 1). Accordingly, overexpression of the agouti protein, an AgRP-like peptide results in the obese, hyperphagic phenotype of the agouti ($A^y$) mouse [47].

The pivotal role of these cells in the acute regulation of feeding in adult mammals has been further substantiated by toxin-mediated AgRP-neuron ablation: ablation of these neurons in mice led to a hypophagic and hypermetabolic phenotype [48,49]. However, this phenotype of AgRP-neuron ablated mice conflicts that of NPY/AgRP single or double KO mice [50], indicating that either developmental compensation accounts for the lack of a phenotype in the latter mice, or alternative neuropeptides or transmitters are released from these cells. Indeed, AgRP neurons are GABAergic and thereby account to a large extent for the GABAergic innervation on neighbouring POMC cells [30] (Fig. 1). Strikingly, acute ablation of AgRP neurons in adult mice causes disinhibition of neurons, as assessed by cFos immunoreactivity, in numerous brain sites including the ARC, PVN, DMH, medial preoptic area, lateral septum, and nucleus of the solitary tract [51]. Consistent with a role for GABA-signaling in AgRP neurons, AgRP-cell restricted inactivation of the vesicular GABA transporter results in leanness [52]. Although the functional significance of disinhibition of individual remote sites awaits experimental proof, it highlights the widespread interconnectivity of feeding-related neurocircuits and particularly the rapidly evolving role for GABAergic transduction in control of feeding [51].

## 3. The melanocortin system as a target of insulin and leptin action

Regardless of the downstream effector neurons in the melanocortin system, the analysis of the hormonal control of POMC and AgRP neurons has progressed immensely. Both neuronal populations express insulin and leptin receptors and are targeted by the respective hormones [14,34,53], and large body of evidence has revealed an important role for leptin-activated STAT3 signaling in the CNS and POMC/AgRP neurons specifically in control of energy homeostasis. Neural specific disruption of STAT3 results in hyperphagia, obesity, diabetes and infertility [54]. Along this line, mice with disruption of STAT3 specifically in leptin receptor expressing neurons develop profound obesity [55]. Further substantiating a role for STAT3 signaling in POMC/AgRP neurons for maintaining normal energy homeostasis, lack of leptin receptor signaling or inactivation of STAT3 specifically in POMC neurons results in obesity and other elements of the metabolic syndrome in rodents [56,57], and deletion of STAT3 from AgRP/NPY neurons results in modest weight gain accompanied by mild hyperphagia. Interestingly, mice with a constitutive activation of STAT3 signaling specifically in AgRP neurons are lean due to increased locomotor activity [58], assigning AgRP neurons and specifically STAT3 signaling in these an unexpected role in control of locomotor activity.

Unlike leptin-mediated changes in STAT3, insulin fails to activate STAT3 signaling and thus alternative intracellular signaling pathways mediating insulin-evoked anorexia via control of the melanocortin system should exist. Since intracellular signaling pathways of both insulin and leptin converge at the level of the PI3 kinase [59], this pathway provided an excellent candidate for convergence of insulin and leptin action. Indeed, blockade of PI3 kinase activation inhibits both insulin's and leptin's acute anorexigenic effect in rats, indicating an important role for PI3 kinase activation in mediating the intracellular effects of both hormones [60,61]. Subsequently, it was demonstrated that insulin and leptin-evoked PI3 kinase activation results in Akt-mediated phosphorylation of the transcription factor FOXO, thus leading to disinhibition of POMC transcription [28,29].

## 4. $K_{ATP}$ channels integrating hormonal and nutrient signaling

Besides regulating gene expression, PI3 kinase signaling and especially its product PIP3 have been shown to regulate cell excitability by binding to the regulatory subunit of ATP-dependent potassium channels ($K_{ATP}$ channels) and this regulation has been demonstrated to also control peripheral glucose metabolism, food intake and energy homeostasis [62] (Fig. 1).

$K_{ATP}$ channels were initially identified as metabolic sensors in the pancreas, where they sense elevated glucose levels through changes in ATP concentrations as a result of glucose metabolism in the cytoplasm. Elevated ATP concentrations lead to the closure of $K_{ATP}$ channels, thereby opening voltage dependent calcium channels, increasing intracellular calcium concentration, and thus triggering vesicle release [63].

$K_{ATP}$ channels are widely expressed in different cell types of the brain: glia cells and neurons of the hippocampus and the hypothalamus [64–66], including POMC/CART- and AgRP/NPY-expressing neurons. PIP3 modulates $K_{ATP}$ channel activity via three different proposed mechanisms: (a) PIP3 increases the probability that $K_{ATP}$ channels are open, which indirectly lowers the ability of ATP to inhibit the channels; (b) PIP3 directly decreases ATP binding to the channel

[62,67,68]; and (c) PIP 3 modulates $K_{ATP}$ channel activity by degradation of local actin filaments around $K_{ATP}$ channels [32]. The proteins that control and induce actin cytoskeletal degradation after PIP3 accumulation have not been identified yet, although Rho or Rac GTPases are likely candidates [69].

Recent studies have addressed the role of PIP3 mediated $K_{ATP}$ channel modulation in hypothalamic neurons with regard to energy and glucose metabolism. Constitutive activation of PIP3 formation in POMC cells by conditional ablation of the phospholipid phosphatase and tensin homolog (PTEN) results in hyperphagia, diet sensitive obesity, and leptin resistance [70]. Moreover, transgenic expression of a mutant subunit that prevents ATP-mediated closure of $K_{ATP}$ channels resulted in impaired whole-body response to a systemic glucose load although with no obvious effect on body weight [71]. What remains a controversial aspect of $K_{ATP}$ channel regulation is the notion that insulin activates these channels both in anorexigenic POMC neurons as well as orexigenic AgRP/NPY-neurons. Thus, the physiological effect is expected to counteract each other via simultaneous silencing of both branches of the melanocortin system. However, further experiments will have to address the signaling threshold for either cell type, to address whether for example differential activation states of signaling pathways such as the PI3 kinase pathway differ between both cell types.

## 5. Peripheral hormones controlling energy homeostasis via non-arcuate sites

However, although POMC cell restricted Pten deficiency and subsequent activation of $K_{ATP}$ channels in POMC neurons resulted in hyperphagia and obesity, constitutive activation of PIP3 formation in all leptin-sensitive neurons reduces adiposity by increasing energy expenditure. Furthermore, these mice demonstrate a brown adipose tissue-like differentiation of white adipose tissue, accompanied by increased uncoupling protein 1 (UCP1) expression, which underlies an enhanced leptin-stimulated sympathetic nerve activity in WAT [72]. These experiments clearly indicate that leptin targets other neurons than POMC- and AgRP-expressing neurons in the ARC and that activating the same intracellular leptin-stimulated signaling pathway in different leptin responsive neurons may result in even opposing effects on energy homeostasis.

In line with the latter notion, leptin receptors are expressed not only in the ARC, but also in the VMH, the LH and other hypothalamic sites, as well as in neurons of the midbrain [35]. Indeed it could be demonstrated that inactivating the leptin receptor in the VMH causes obesity, which is exaggerated when leptin receptor deficiency is present both in POMC and VMH neurons [73].

Furthermore, expression of the leptin receptor and its functional signaling in dopaminergic neurons of the ventral tegmental area (VTA) provided an interesting link between the hormonal control of food intake and the reward system (Fig. 2) [74]. Similarly not only leptin but also insulin targets the dopaminergic ventral tegmental area neurons [75–77]. Consistent with this, performance of an animal in tasks with food as a rewarding substance is enhanced by food restriction or deprivation, states in which insulin and leptin levels are low [78–81]. Thus, these experiments clearly indicate that insulin and leptin both within and outside the hypothalamus target other neurons, which control important brain functions associated with feeding such as control of reward.

## 6. Peripheral hormonal signals in food reward

Neuronal mechanisms underlying food reward may be closely related or similar to mechanisms of drug reward. Food and water are naturally rewarding and motivating stimuli, which promote burst firing of dopamine neurons, thereby facilitating goal-directed behavior towards the acquisition of food or water [82–85]. The brain

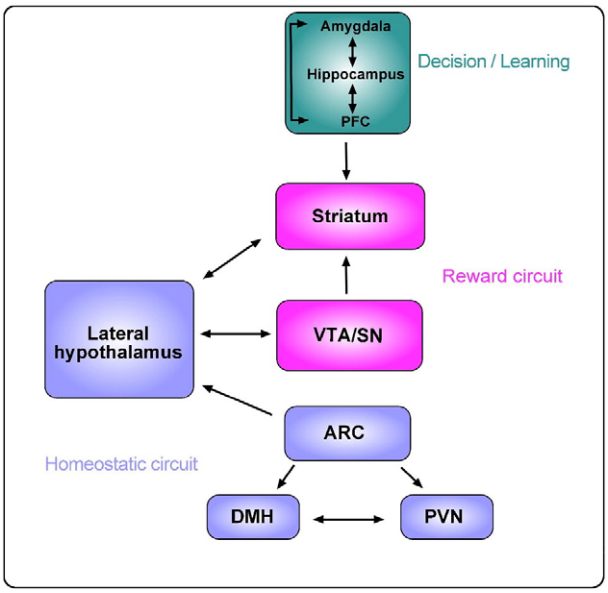



**Fig. 2.** Integration of neuronal pathways involved in reward and regulation of energy balance. Abbreviations: ARC: arcuate nucleus; DMH: dorsomedial hypothalamus; PFC: prefrontal cortex; PVN: paraventricular nucleus; SN: substantia nigra; VTA: ventral tegmental area.

regions activated by a food reward stimulus (e.g. highly palatable food) are the same regions responsive to drugs of abuse and involved in the development of drug addiction [86,87]. There is also evidence that dysregulation of those drug reward circuits is involved in the development of obesity. Imaging studies in humans demonstrated that dopamine D2 receptor availability is decreased in obese individuals in proportion to their body mass index [88], and treatment of *ob/ob* mice with dopamine D1/D2 receptor agonists normalizes hyperphagia, body weight gain, hyperglycemia, and hyperlipidemia [89]. Notably, chronic food restriction and maintenance of low body weight increase the self-administration and motor-activating effects of abused drugs [90–92]. By using a lateral hypothalamic self-stimulation rate-frequency method, Carr could demonstrate that food restriction augments the rewarding effect of diverse drugs of abuse [93]. Electrical self-stimulation of a subset of neurons in the LH results in an induced food intake in rats and this effect is augmented with food restriction [94]. Furthermore, self-administration of food, and the ability of food to condition an animal to a place preference are increased with food deprivation [78–81].

There is evidence that peripheral hormonal signals, such as insulin and leptin, modulate the reward system of the brain. As indicated above, food restriction and fasting, states in which insulin and leptin levels are decreased, enhance the rewarding and reinforcing properties of drugs. Both insulin and leptin, when administered intracerebroventricularly (icv), are able to reverse the shifted threshold of LH self-stimulation after food restriction [95,96]. Moreover, icv-administered leptin attenuates the acute food deprivation-induced relapse to heroin self-administration in rats [97]. Furthermore, peripheral applied leptin reverses sucrose-conditioned place preference in food-restricted rats [98]. Finally, studies have implicated the IRS2-Akt signaling pathway as an important regulator of dopamine cell morphology and opiate reward [99]. Thus, insulin and leptin may play a role in modulating the reward system in the CNS.

Additionally, there is much evidence that insulin and leptin are directly involved in the food reward system of the brain. Activity of the dopaminergic midbrain system is known to be involved in motivational driven behaviors, and the aspect of reward or motivation can be tested in different behavioral paradigms. Figlewicz and colleagues analysed the ability of icv-administered insulin and leptin at physiological doses, which do not cause a change in body weight or

food intake, to modify performance of an animal in such food reward tasks in *ad libitum* fed rats. To evaluate whether insulin is able to alter performance of a sucrose-lick task, which is known to be dependent on intact dopaminergic signaling, the authors tested the effect of icv insulin and the D2 receptor antagonist raclopride. Although ineffective on its own, icv insulin combined with raclopride significantly suppressed sucrose-lick rate in a five-minute task [100], indicating that insulin indeed could decrease the reward value of a sucrose solution. The conditioned place preference (CPP) task, which also depends on intact dopaminergic signaling in the CNS [78,98], evaluates the preferences for environmental stimuli that have been associated with a reward (food or drug). CPP to a high-fat diet is blocked in *ad libitum* fed rats given icv insulin or leptin [101], again indicating that insulin and leptin can influence the reward value of food. Finally, in a sucrose self-administration task, in which the motivation to work for a sucrose reward is measured by pressing a lever, both insulin and leptin icv decreased the number of presses on the lever [102]. These results indicate that both insulin and leptin are involved in different aspects of food reward: hedonics, place preference, and the motivation to work for a reward [103].

Consistent with a modulatory role of these hormones on the reward system, insulin and leptin receptors, as assessed by double-labeling fluorescence immunohistochemistry, are co-expressed with tyrosine hydroxylase – a marker for dopamine neurons – in the VTA and substantia nigra [104]. Furthermore, as previously shown for the mediobasal hypothalamus [59,105], insulin or leptin administration into the VTA results in an increased formation of PIP3, as compared to controls injected with aCSF [103]. PIP3, the product of the PI3 kinase serves as an indicator for a functional signaling pathway. Furthermore, it has been shown that leptin receptor mRNA is expressed in the VTA and in response to peripheral, icv or direct VTA leptin injection the intracellular JAK–STAT pathway becomes activated [74,96]. This evidence suggests that the insulin and leptin receptors are functional, and that the respective signaling pathways are indeed activated in these dopaminergic cells. Moreover, activation of the JAK–STAT pathway by leptin has been shown to be critical for the effect of leptin in the VTA to decrease food intake [74]. Finally, electrophysiology studies revealed that leptin significantly reduces the firing rate of dopaminergic neurons in the VTA [74].

Figlewicz and colleagues [108] have identified an additional potential target for insulin action on dopaminergic neurons: the dopamine transporter (DAT), which provides the primary mechanism through which extracellular dopamine is cleared from the synapses, thereby regulating the magnitude and duration of dopaminergic signaling [106]. Insulin signaling through the PI3 kinase- and Akt-pathway promotes cell-surface expression of the DAT *in vitro* [107]. Chronic insulin icv treatment results in an increased DAT mRNA expression in the VTA/substantia nigra and, consistent with this, food deprivation results in a decreased expression and activity of the DAT [108,109].

In a recent study, the efficiency of insulin to decrease intake of palatable food when administered acutely in several CNS location was compared in *ad libitum* fed rats. Acute insulin injection directly into the ARC decreased sucrose self-administration [110]. Surprisingly, insulin had no effect on sucrose self-administration when injected in the VTA or the nucleus accumbens (NAc). Taken together, these results suggest that insulin's effect on sucrose self-administration occurs primarily at the ARC and indicates that the VTA and NAc circuitry is rather engaged via multisynaptic activation [110]. However, assessment of the functional role of insulin receptors expressed on VTA neurons still awaits further direct experimental investigation. Nevertheless, it appears possible that both insulin and leptin receptors expressed on VTA neurons cooperate with presynaptic actions of these hormones to mount the full regulatory response elicited to control energy homeostasis.

Expanding our knowledge of how peripheral hormones regulate energy homeostasis and how these signals are integrated in multiple brain circuits is pivotal for a better understanding of the pathomechanisms responsible for the growing obesity epidemic. In western populations, very little efffort is necessary to procure high caloric food and ingestion of highly palatable food takes place even if there is no metabolic need for it. Motivation about when to eat and more importantly when to stop eating is clearly a more complex system with other brain regions involved additional to the well known homeostatic system of the brain. The use of pharmacological studies as well as conditional gene targeting will ultimately allow to understand how the homeostatic and the cognitive systems of the brain are connected, which is a prerequisite for the development of novel therapeutic approaches for the treatment of type 2 diabetes mellitus and obesity.

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